Assignment Predictions for Machine Reassignment Problem

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In partial fulfillment of the requirements for the degree of

MSc in Computer Science (Artificial Intelligence - Online)

2025

DECLARATION I, Piotr Bielski, hereby declare that this thesis, titled "Solutions Analysis to Machine Reassingnment Problem", and the work presented in it are entirely my own except where explicitly stated otherwise in the text, and that this work has not been previously submitted, in part or whole, to any university or institution for any degree, diploma, or other qualification.

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Abstract

In the era of Big Data, single-GPU and CPU setups for machine learning tasks are becoming insufficient in the computational power needed to keep up with the fast rates of increasingly large data volumes generated by the scientific communities and private industries [1] [2]. The use of cloud computing and high-performance computing platforms is becoming more widespread as the coupling of AI with more available hardware can significantly speed up the training phase. However, growing demand for computational resources has a negative impact on the environment due to the high consumption of electrical power. Therefore, it is critical such services run as optimally as possible. Efficient resource management techniques can not only reduce operational costs but also reduce the environmental impacts. This report aims to investigate challenges in efficient resource management, solutions to the ROADEF12 Machine Reassignment problem, and attempt to predict optimal process reassignments based on common patterns.

Keywords: Machine Reassignment, Resource Management, Machine Learning, Optimization, Efficiecy

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Chapter 1

Introduction

1.1 Resource Management in Computing

High-performance computing platforms, cloud services, and data centers provide computational resources to a wide range of resource-intensive scientific and industrial applications. However, our growing reliance and demand on these platforms and services results in an enormous amounts of electrical power being comsumed. [3] Moreover, the number of these computing instances is ever-increasing and constitute a significant portion of global energy usage, leading to more CO2 emissions and have a direct impact on the environment. [4] Therefore, operational inefficiencies not only have and impact on our environment but also increase the operational running costs. Further issues with resource management inefficiencies are pointed out by (Dillon), stating cloud providers stuggle to maintain their SLA and is becoming a major issue. [5] Considerable strategies have already been implemented to improve energy efficiency, however, there is still potential for operational and technological optimizations that requires less captial investment but can yield significant results. [6]

1.2 Overview of Google MRP

In this context, Google proposed a problem for the ROADEF/EURO challenge in 2012 (https://www.roadef.org/challenge/2012/en/index.php), presenting a complex and large-scale machine reassignment problem (MRP), where a set of processes are assigned to a set of machines have to be reassigned to improve the overall efficiency of the system, while balancing machine usage improvement and moving costs, under resource and operational constraints.

1.3 Motivation

The goal of this report is to investigate the initial and final assignments of processes from solutions submitted to the ROADEF12 challange. The idea is to identify common patterns and determine whether, given these insights, it may be possible to predict optimal process reassignments without performing a search. Furthermore, once common patterns are established, a model is to be developed to use these rules to complete the machine reassignment problem. The contribution from the outcome of this report is to aid in the research for more efficient resource management solutions that improve on operational inefficiencies and reduce our carbon footprint.

1.4 Research Questions

- RQ: Are there common strategies, patterns, or techniques in the solutions?
- RQ: What were the optimizations to produce the best solutions?
- RQ: Can we use these common patterns to make predictions for optimal reassignments?

1.5 Thesis Structure

Chapter 1 presents the introductory information discussed in further detail later and outlines the research questions.
Chapter 2 provides background on the reasoning behind the research topic, the organizations involved in organizing and promoting collaborations on such problems, the problem definition itself.
Chapter 3 discusses the research objectives in more detail, the research methods applied when searching for related works, and the review of literature.
Futher Chapters will be added later.

Chapter 2

Background

2.1 Effects of Experimental Optimizations

Successful implementations of highly optimized and scalable systems have shown to drastically reducing the time-to-insight, even over larger datasets, resulting in highly accurate and ready-to-use AI models. In the domain of physical sciences, researchers [7] were able to achieve massive performance increase by deploying a highly optimized deep learning algorithm and achieving state-of-the-art classification accuracy in just 8 minutes using 64 K80 GPUs on the Cooley supercomputer, with distributed training using Horovod, and open-source tools such as Keras and TensorFlow, as compared to their previous result of 5 hours using single P100 GPU without distributed training. Authors conclude that achieving highly scalable AI model to larger data and feature sets is indeed possible however requires considerable amounts of experimental work and human effort to fine tune the model and training pipeline. Futhermore, another highly scalable deep learning training system was developed and trained on the ImageNet dataset in under 7 minutes using 2048 of the less powerful P40 GPUs as compared to the state-of-the-art GPU-based system consisting of 1024 P100 GPUs that took 15 minutes.

The authors focus was on optimizing batch sizes to improve system scalability by reducing communication-to-computation ratio and point out the communication steps usually becomes the bottleneck of the system with a large number of GPUs, thus presenting difficulties in near-linear scalability with increasingly large clusters. [8] These examples highlight the imporantance and difference between highly optimized systems compared to more general, inefficient approaches and emphasize on the need for standardized guidelines due to the significant experimental component in fine-tuning models.

2.2 ROADEF/EURO Contests

The ROADEF/EURO contest is jointly organized by French Operational Research and Decision Aid Society (ROADEF) and Association of European Operational Research Socity (EURO) and appears regularly since 1999.[1] The challenges are open to everyone and aim to promote innovation and collaboration between researchers. Generally, participants are split into two categories; (1) Junior category for young researchers, and (2) Senior category for experienced and qualified researchers. The challenges provide an opportunity to tackle complex industrial optimization problems and have been received as widely successful. [9] As for the specific challenge investigated in this paper, 82 teams registered for the contest and needed to submit solutions for qualification to be evaulated on dataset ”A”. 48 teams submitted solutions and only 30 qualified. The qualified teams then moved to more realistic set of instances made available to them, called dataset ”B”, to adjust their programs and 27 teams submitted an updated version to be evaluated against the final unknown dataset ”X”.

[1]https://www.roadef.org/challenge/2022/en/

2.3 Google Machine Reassignment Problem

Google proposed a problem for the ROADEF/EURO 2012 challenge on on the topic of resource management, called the Machine Reassignment Problem (MRP). MRP is an optimization problem, commonly encoutered in areas such as cloud computing, high-performance computing platforms, data centers, and relates to the efficient management of compute resources. The overarching goal of the problem is to make the most optimal use of resources given a set of tasks to be completed across a distributed set of computing resources.

2.3.1 Decision Variables

Each machine has limited resources such as RAM, CPU, DISK, and each process consumes different amount of each resource. A solution involves reassigning processes across the set of available machines in a way that optimizes for cost-related objectives and adheres to constraints.

- **Machines**: Refers to set of machines, $\mathcal{M}$, used for the execution of processes. Each machine, $m \in \mathcal{M}$, has a set of characteristics such as CPU, RAM, DISK.
- **Processes**: These are the set of processes, $\mathcal{P}$, that need to be executed by being assigned to a machine, $\mathcal{M}(p) = m$. Each process, $p \in \mathcal{P}$, has specific resource requirements (CPU, RAM, Disk) and may have constraints or dependencies with other processes.
- **Initial Assignment**: All processes are initially assigned in a way that may not be optimal, but to satisfy constraint requirements. The initial assignments are denoted as $\mathcal{M}_0(p)$.

2.3.2 Constraints

Constraints define a set of limitations to which a solution must adhere and reflect physical restrictions such as available compute resources, process conflicts, dependencies, and migration of processes over distributed network.

- **Capacity Constraints**: A process cannot be assigned to a machine if the machine does not have enough available resources, $\mathcal{R}$. A mechine must have enough capacity of all resources (CPU, RAM, DISK) to host the process. The capacity of resource is defined as $\mathcal{C}(m, r)$.

- **Conflict Constraints**: A set of processes form a service, $s \in \mathcal{S}$. The conflict constraint defines that processes belonging to same service must run on distince machines to avoid conflicts, and does not allow processes of the same service to run on the same machine.

- **Spread Constraints**: Similarly, a set of machines form a location, $l \in \mathcal{L}$. The spread conflict defines the minimum number of locations, *minSpread*, over which processes of a service must be distributed.

- **Dependency Constraints**: Services can have a dependency on another service. In such cases, processes of the dependant service must run in the neighbourhood of the other service processes. Here, a neighbourhood is defined as a set of machines as well, $n \in \mathcal{N}$.

- **Transient Usage Constraints**: While a process is migrated from one machine, m, to another, m', the resources required by the process are temporarily consumed on both machines. Eventually, once the migration is complete, the resources are freed from the initial machine.

2.3.3 Objectives

Objectives are the variables that need to be optimized and relate to costs such as the cost of running a process on a specific machine and the cost of evenly balancing processes across machines.

- **Load Cost**: Each machine has a safety capacity per resource, $\mathcal{SC}(m, r)$, over which a penalty is incurred for over-utilizing machine resources beyond safe threshold.
- **Balance Cost**: This objective aims to ensure there is a balance of available resources across machines, useful for future process assignments. It incurs a pentalty for imbalances, for example, for a machine having 100 CPU usage but 0 RAM usage as it essentially renders RAM unusable.
- **Process Move Cost**: Accounts for inconvenience and potential downtime associated with migrating processes and incurs a cost for moving a process to another machine.
- **Service Move Cost**: Designated to balance the number of moved processes across different services. It is defined as maximum number of moved processes for any single service.
- **Machine Move Cost**: Cost of migrating any process to another machines. The cost varies depending on specific source and destination machines.
- **Total Objective Cost**: Weighted sum of all individual costs.

To restate again, the objective of the problem is to minimize the **Total Objective Cost** while adhering to the **constraints**.

Chapter 3

Related Work

3.1 Research Objectives

The research objectives to be accomplished through literature review are to gain insights into the different approaches taken by the submitted solutions for the MRP, investigate whether they were known or novel techniques, whether the solutions prioritised specific actions over others, such as preference to initally migrate large processes over small processes, and whether any sort of randomization played crucial role in their success. Should common patterns be found, they could provide insights to develop strategies for the reassagnment of processes without having to perform the actual search for the best way.

3.2 Research Method

The literature research was performed primarily on Google Scholar, NUIG James Hardiman Library, and Scopus for documents containing keywords: "Machine Reassignment Problem", "Google Challenge 2012", "Resource Management", "ROADEF12" and "Resource Optimization". Abstract sections of the resulting documents from the search queries were scanned and selected for in-depth analysis depending on their relevance to the topic. The query was further narrowed down to include peer-reviewed papers in high-quality journals (Q1 and Q2) that were published from the year of the publishment of the Google Machine Reassignment Problem, 2012 and onwards. Furthermore, of the selected relevant papers, their references and citations were investigated to provide more informational background and potential implications on the research that followed.

3.3 Literature Review

3.3.1 Solutions Proposed for the MRP

Solutions were evaluated on each instance by the gap of the team's cost from the best known cost divided by cost of reference solution, where the reference solution is where no process is reassigned.

The winning team, S41, proposed a local search algorithm. The quality of the search was evaluated using lower bound based on load and balance costs and exploring four neighbourboods: (1) Shift, (2) Swap, (3) Big Process Rearrangement, and (4) Chain Shift. Their solution found two best known solutions (BKS) on dataset X and three BKS on dataset B. [10]

Second place team, S38, present a Constraint Programming (CP) model with Large Neighbourhood Seach (LNS). They found one BKS on dataset B and two on dataset X. [11]

In the third place, team J12 proposed a solution of two components; (1) LNS using Mixed Integer Programming (MIP), and (2) fast greedy hill climber. They found one BKS on dataset B. [12]

Fourth place team, J25, proposed LNS approach with CP to iterate over improving solutions. The authors deemed the Service Move Cost insignificant and were able to save computing time by excluding if from the search process. [13]

3.3.2 Patterns in Proposed Solutions

Most common components of the top 5 solutions is a mix of Lower Bounds, Linear Search, Large Neighbourhood Search, and Constraint Progreamming.

Chapter 4

Data

Chapter 5

Methodology

Chapter 6

Experiments

Chapter 7

Results

Chapter 8

Conclusion

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